

ActInf GuestStream 046.1 ~ Denise Holt, "Active Inference AI & the Spatial Web"

Source: [YouTube](#) Playlist: GuestStreams Duration: 1:18:59 | Uploaded: Unknown Transcript method: auto_caption

hello it's June 23 2023 we're in active guest stream 46.1 with Denise Holt so Denise thank you very much for joining we'll have your presentation first any length of time you want and then we'll have a discussion about some of the topics you're bringing up and I'll read some comments from the live chat too so thanks again Denise looking forward to your presentation hi Daniel thank you so much for having me today um yeah so let's just Dive Right In today I'm going to be talking about active inference AI and the spatial web and uh for those who are not familiar um there is a company called versus AI and they're basically uh delivering the next generation of AI this is beyond uh the current machine learning deep learning models that we're all so familiar with that are incredible tools but really those are tools they are great at creating content performing tasks things like that but you know they operate in a way that is um you know pattern matching and they're all siled in their ability for what they they can do they're trained on enormous amounts of data to increase the pattern recognition you know and hopefully increase accurate you know outputs but that's really what they do um what versus has created is something completely different um and it's based on active inference and so active inference AI is entire really new and this is the AI that will be able to give us to uh AGI you know leading to synthetic self-evolving intelligence so today um what I'm going to be talking about in uh this deck is um you know active inference AI and the spatial web and how these Technologies are going to affect and change the world we'll talk about how active inference Works within with artificial uh how the spatial web protocol enables the perception and the belief updating of the AI and how these Technologies together enable the only AI in the world that can run critical operations um for systems or infrastructure smart cities you know the planet so when you're talking about airports hospitals things like that um so yeah so let's Dive Right In so the spatial web uh what is that so the spatial webs being called the network of everything basically it's the next evolution of the internet and it's powered by AI so this is taking us from a library of pages and documents that we have in the world wide web

and um oh my gosh my dog is wanting out I'm so sorry I have to let him out sorry about that so um the next evolution of the internet um now you know as I said World Wide Web a library of pages documents um you know everything is static um the spatial web protocol hstp hyperspace transaction protocol and uh the programming language for the protocol called uh hsm1 which is hyperspace modeling language this new protocol takes us into a library of spaces where you know we have spatial domains and everything within any any space can be locatable and programmable within this new network so and it's just extending the protocol that we have now right so same internet just more capabilities now um Dr Carl J friston and the versus AI white paper and I'm sure a lot of you are familiar with him uh the father of active inference and the free energy principle now um so versus with uh with Dr friston has developed a new type of AI that mimics the self-organized systems of nested intelligence found in nature and it's based on his methodologies of active inference and the free energy principle and the spatial web protocols which I just mentioned which were developed by versus AI um in December they put out a white paper called designing ecosystems of intelligence from first principles and just to back up a second you know versus created the protocol uh the hstp and the hsm1 but donated it to the public because nobody can own the internet but they needed that in place for what they wanted to do as a company so that framework needed to be there so then they also donated the protocol and the IP to the IEEE which is one of the largest core standards bodies that develops everything with you know electronics and Engineering you know they're responsible for the core standards around things like Bluetooth and Wi-Fi and and things like that so the core standards have been being developed around the protocol for the last almost three years and um and yeah versus in that time has been building you know what they've what they've been building which I'll I'll be talking about so what is a Swype paper about um it's basically about shared intelligence so in the white paper uh they proposed cyber physical nested ecosystems of distributed intelligence joining humans machines and AI agents on a common Network so um we're talking about uh humans as integral participants uh adaptive Behavior self-evidencing self-organization belief updating over several scales um belief sharing over ensembles of Agents so we're talking about a network of intelligent agents and collective intelligence and then of course the protocols uh hstp and hsm1 the knowledge graph and digital twins um digital twins of our planet and all nested systems and entities within them um so nested ecosystems intelligent agents both human and synthetic sensing and perceiving continuously in evolving environments making sense of changes updating their mental model and what they know to be

true and then acting on the new information they receive and in the white paper it's also discussing a network of distributed intelligence so this enables a cognitive architecture made up of the collective intelligence of multiple agents that continuously communicate coordinate and collaborate with each other um individual and specialized intelligences are all coming together on uh they're speaking a Common Language hsm1 and hstp it's efficient powerful cross communication to perform tasks regulate systems and address problems in real time and it scales up and grows in tandem with humans so active inference and the free energy so what exactly is active inference active inference is a biologically inspired approach to AI systems as a method for understanding Behavior incorporating the brain cognition and behavior modeled after the design principles from nature and from how the brain nervous system and the body act and react uh one of the questions that Dr friston asks in this paper that is uh an important question in reference to active inference is you know with AI systems you need to be able to understand what is it that intelligence must be given that intelligent systems exist in a universe like ours so active inference is a theoretical framework that describes how agents like people animals robots or artificial systems can maintain their internal States and behavior in unison with their objectives or goals so what is the free energy principle the free energy principle is the theoretical Foundation of active inference it proposes that organisms or agents maintain their internal States and behavior by acting in a way that minimizes the difference between their current beliefs about the world and what they expect to be true so in other words the agents try to make things more predictable by mitigating the difference between what they expect to happen and what actually ends up happening and they do this uh by way of a cons a continuous cycle of improving their perception by acting on the environment um so basically it helps to extract the signal from the noise and extract the regularities from the irregularities um so extracting the prediction error minimizing the prediction error and you know there are there are two parts to this uh to this inference cycle you know there's the perceptual inference part uh which minimizes the free energy by improving the internal model to better match the sensory uh input um making Bayesian inferences about the world the brain minimizes free energy by optimizing perception so through hypothesizing and observation in other words the brain models the environment better improving the perception now the active inference part uh minimizes the free energy by acting on the environment so acting on the environment to reduce the surprises the disorder or the unpredictability and action can indirectly influence the observation so active inference the active inference part of free energy minimization is equal to lowering the surprise in the sensory observations by acting on the environment actions can't

directly change observations but they can indirectly change them by acting on the environmental States so then we get to active inference AI so in the context of artificial intelligence active inference and the free energy principle explain how agents can learn and adapt to new situations and how they can generate predictions and plan actions based on their goals or objectives which leads to more accurate predictions and results which is the real goal of AI active inference AI enables us to overcome current limitations with machine learning and deep learning AI models providing a realistic path towards artificial general intelligence um so you know the idea is that an agent can use active inference to minimize the free energy and optimize Its Behavior in a way that's consistent with its objectives which uh yeah which leads to more accurate AI so the action perception Loop um the perceptual inference and active inference unfold continuously and simultaneously uh underlying deep continuity between perception and action they're both sides of the same coin performing the same free energy minimization algorithm so the free energy can be minimized by improving the perception and then acting on the environment in this Loop that just keeps narrowing down the focus and then also by learning the generative model you know how the brain codes the um so you know that's the side that the the spatial web protocol actually helps with because it bakes context into every person place or thing in any space so that context informs the um the AI so then it it's optimizing the expected precision and regulating the learning so thus we update the model over and over to improve our understanding of the world and that is what the AI does as well so how does it do this so self-optimization and self-evolution so active inference Ai and the free energy principle through the Bayesian inference allows a neural network to self-optimize through intake and continuous updating of new real-time sensory data while simultaneously considering previously established outputs and determinations so past decisions plus the new input uh lead to Future outcomes um and that that cycle is a self-optimization and self-evolution cycle so a self-evolving system evolves over time and current systems current AI systems are more like a machine they're just neural Nets that are trained to do a specific task but a self-evolving system is learning from moment to moment and upgrading its World model and thus it mimics biology while also enabling general intelligence So within the free energy principle neural networks self-optimize through a set of mathematical rules enabling Next Generation artificial intelligence to efficiently learn predict plan and make decisions spatial web protocol let's talk about this so what is the spatial web protocol hstp hyperspace transaction protocol it gives browsers the ability to link spaces with an ID for every person place or thing both digital and physical so virtual or real but it's much

more than a location identifier hstp is also a gatekeeper and allows various parties the ability to agree on who what and where anything is in space who owns it has access to it and what can be done with it so hstp gives browsers the ability to link the spaces with an ID oh did I get this in here so hstp is multi-dimensional query so it allows for query over multiple Dimensions identifying localizing and updating the attributes conditions excuse me contingencies and interrelationships of objects in space and over time so if it can be defined and calculated in the sphere of natural or Digital Science it can be searchable through hstp so hsml this is the programming language of uh that informs the protocol and I like to call it the smartest contract around because it's it's the foundational contract now for every person place or thing in any space in any reality so the only way to create a truly technologically augmented existence is to be able to consider and measure the contextual elements that affect the expression of shared information by and between all objects in space uh this is known as computable context and this is what hsml hyperspace modeling language was made for and what are the some of the uh contextual elements that can be programmed location the where the when of anything in any space uh and you know you're talking about you know reality the different realities spaces time and channels as well and then activities so the the what and the how you know and this is this is in regards to things like rights and credentials or claims or activities and then when you're talking about identities uh The Who and the what of anything in any space you know you're talking about Authority or domains or users or assets so all of these uh all of these uh contextual elements can affect the expression of the shared information by in between anything so belief updating through context and sensors so hsml is a cipher for context the spatial web which is web 3.0 the next evolution of the internet is library of spaces that contain objects people places and things these objects do things and change over the context or circumstances that govern these shifts and changes is the most important factor to consider if we're to understand how objects relate to each other to people and to their environments hmm so the free energy principle and hsml the free energy principle is perfectly suited to the programming language of the spatial web hsml uh the hyperspace modeling language which enables computable context to context based on defining recording and tracing the changing details in physical and digital Dimensions social Dimensions uh meanings culture conditions circumstances situations whether geometrical geopolitical or geosocial by Nature self-evidencing and belief updating so hsml computes context enabling the ai's perception to understand the real-time Changing State of anything in the world accumulating evidence for a generative model of one's sensed World also known as self-evidencing we're talking about

multiple agents so individual intelligences and based on each agent's generative World model unique perspective frame of reference and nested ecosystems of these intelligences uh and different levels of self-organization so all of this results in a collective shared intelligence so uh pollen architecture and Markov blankets so the spatial web uh is a hollonic structure so active inference uh well active inference AI within the spatial web on the versus cosm OS platform operates with a hollonic structure of nested spatial domain domains so if you're not familiar with what a hollon is a hollon is where you know something can be a hole in and of itself and yet also be a part of something greater so the human heart is a great example the human hearts made of cells each one of those cells their whole cells in and of themselves yet they're also part of the heart and then the heart is a whole heart but yet it's part of the human body so you have this nested structure of whole parts that are governed by the outside part but still have this intrinsic governing going on within itself as well so each component is a whole element part of the entire organism uh their nested entities and they're governed by the rules of the greater influencing organism and then uh there's they have inherited uh they inherit the governing and it's passed to the internal parts that are nested within it and each part has its own governmental considerations for its specific intrinsic requirements as well so this is how the spatial web is set up to work as well so you know when you're talking about spatial domains you're um you know like for instance you can have a um you know a restaurant that's inside of a skyscraper building that's inside of a city and that city is inside of a uh State you know these nested domains and then the active inference AI also uh includes the principle of Markov blankets at each level of analysis so the mark off blankets uh Define define boundaries acting as partitions to mediate the interactions between the internal and external States um of of the spatial domains such as like a single unit or a region or entire you know complex Network and this results in a self-organizing system and a Markov blanket is just a mediator between how things are attuned to each other through the free energy principle so nested domains in the spatial web so one good example of how this may work uh is you know if you take a factory right so you know a factory is a part of the supply chain and uh has raw raw materials feeding into it from various locations like mines or you know anywhere and then the factory produces finished goods that may ship to another Don another destination like a uh Warehouse right like a distribution warehouse so just as that factory is an object within a larger ecosystem of the supply chain every object within the factory belongs to its internal ecosystem right so with a factory you've got all the internal ecosystem of what's going on in that factory but yet it's still part of this Global supply chain with you know um uh you know

raw materials being fed into it and then it when it's done with what it's doing it's feeding out to other like distribution um you know Pathways so um you have an internal and an external ecosystem and they simultaneously operate according to the individual internal and external standards the interdependencies the in the inner relationships so you know this is how you can see things working within the spatial web so every object within the entire spatial web network is part of an interrelated ecosystem that feeds a continuous stream of real-time context between all points in space and time this continual interaction and communication stream takes note of all the nuances and changes within the relationships between all objects and the parameters that govern them foreign which leads to Adaptive collective intelligence automation so you have programmable spaces and uh you know within these spaces everything within these spaces have a digital ID and the way the spatial web is set up is it enables zero knowledge proofs between all of these entities so anything inside of any space is uniquely identifiable and programmable with a digital twin of the earth producing a model for data normalization so by programming context into everything in any space it creates this digital twin of the space so the spatial web becomes a digital twin of everything from the planet to every single system every single object within it that's on the network the contingencies and changing details inherent qualities circumstances for all objects and situations can now be measured and computed uh providing a basis for the AI perception affecting all entities and their interrelationships uh to each other so you're talking about adaptive intelligence automation security through geo-encoded governance multi-network interoperability and it enables all Smart Technologies to function together on a unified system so you know um when you consider all of the web 3 Technologies all of the extended reality Technologies you know AR VR um distributed Ledger Technologies so you have all of these technologies that are siled and disparate this brings all of those Technologies onto a common unified Network now with a Common Language between them so it makes all of these all of these Technologies then become interoperable and when you're talking about Smart Technologies when you're talking about like The Internet of Things um there's been no common language for these internet of things to talk to each other and communicate so the spatial web brings all of the internet of things onto the common Network now where they become baked into this um this network and informing the AI so all of the data being gathered by all of these sensors or cameras or you know the context that's baked into the protocol this is all the data that is now coming together that is continually updating in real time and informing the AI so you have ecosystems of nested intelligence the design of intelligence systems must

begin from the physicality of information and its processing at every scale or level of organization. So within the spatial web you have AI that scales up the way nature does aggregating locally contextualized knowledge bases and acting across ecosystems and this maximizes efficiency so this is the complete opposite of the way the machine learning models work because they're top down it's just massive amounts of data and you know that's what's training the machine model AI um but there's a cutoff date so then that model now can only reference this historical data that it's been trained on and you know again it's just a pattern recognition machine the algorithms are just to make sense of the data that it's been fed and how to recognize patterns within it um every time those models have to compute and try to perform an output it requires massive amounts of energy because it still has to pull from its library this massive library to try to make sense of things now the way that the spatial web works with the active inference AI is you have any amount of data you know small amounts of data can now be made smart because you have context informing the AI and you have sensors that are giving it real-time data inputs uh to let it kind of look outward and see what's happening in the world now not you know referencing historical data so this leads to minimizing complexity so when you're talking about the machine learning versus the active inference um for efficiency and accuracy the more complex a system is the more energy it consumes so that's what I was just referring to with all this big data approach on these machine models active inference in the free energy principle naturally minimize the complexity that's the whole idea with the free energy principle is to minimize the prediction error and use the information that it can sense right so it's taking in this the senses from from sensors like the internet of things and cameras and things like that but it's also taking in what it knows about its uh its you know the data at hand right and it's informed protocol the spatial web protocol with all of the context all of the all of the attributes around everything in any space or any data set so you know sensor data is noisy and ambiguous but hsm1 provides clarity with specific and precise contextual data to narrow the complexity and close the gap on the free energy so more certainty means less noise and that each that equals free energy is reduced faster and lower so it's a faster way to operate more efficient and you know another point to this which is really interesting too is that you know most of the data that exists exists behind uh password walls right behind gated uh within gated password areas right whether it's you know on the internet and it's all the data that people have behind a password or whether you're talking about like an Enterprise organization that has all their proprietary data that's internal you know so um the awesome thing about the spatial web is that um and

the active inference AI is that you can be within this network and still have all of your proprietary data gated off because the protocol allows for that security you know it's a it's gated at every touch point so you can put boundaries around your particular data but still be able to access the the the power of the network and the power of the active inference AI now for your proprietary data because it can take that small amount of data and make it smart so um this is a game changer for you know access to AI for any systems in any databases so embodied AI active inference uh is mimics biological design uh and it's so it's like embodied AI with the ability to take action and this is the core engine of the spatial web so what makes active inference so accurate active inference is so accurate because it continually looks outward into the world measuring the world in real time through a Global Network of sensors iot devices cameras robots drones anything that's connected within the spatial web the digital twin network of the world hsm1 informs the AI with precise data extracts the signal from the noise minimizing the prediction error and this mimics the way humans and animals make so the perception of the world in the spatial web active inference AI uh you know create this creates a cybernetic feedback loop of perception of the world and it updates the model of the world with the belief of what it knows to be true gaining the understanding of the intricacies and inner workings of the world so that it can make decisions and take action okay and it provides ever ever evolving intelligence so the more this feedback loop plays out the AI learns more about the world and the results of the actions taken just as a child learns about the world as it grows and interacts with it and the more accurate this AI becomes by further updating its understanding of the world so um it's been described to me that you know when versus launches their cosm operating system and then you know which is going to be uh later this year uh it's going to enable everybody to be able to build these intelligent agents on top of the network right there they have a intelligent app store that's going to be launching with the operating system so then everybody can make these intelligent agents the difference is these agents so when you build an app on say like the Apple Store app store or Google Play You're Building a siled piece of software right those apps are not aware of each other they just do whatever their function is to do the difference is these intelligent agents that you're building on cosm are going to be aware of each other and aware of the network so they're going to be empowered with the AI and be able to act as an agent within the the network now it's been described to me that you know when people are first building these a these you know intelligent agent apps um it's not going to be like this Stark difference of like the power of what it can do versus the power of what these machine models can do because they're all

pretty it's all pretty impressive right now but the difference is is if you take like uh the comparison of like a chimpanzee and a toddler at first they seem about the same intellectually but the difference is that toddler is going to grow it's going to grow into a full fledged adult it's going to continue learning whereas that chimpanzee has maxed out so um so that's going to be really interesting to watch over the next you know a year or two is that we're gonna see this active inference AI really modeling the same way that humans learn and it's going to grow in intelligence with with this you know scores of these intelligent agents now all working together within the network so now one of the really awesome things about this too is that the active inference AI is explainable auditable and governable so versus AI uh just published a groundbreaking industry report about a week week and a half ago called designing explainable artificial intelligence with active inference a framework for transparent introspection and decision making so active inference is explainable AI now how how can this be well active inference can self-report so machine learning models they're black boxes uh you know and and therefore they can't quantify their uncertainty they're unobservable unalterable and unknowable basically the the way the algorithm is processing the data and coming up with its output you you can't see what's happening you can't know how it's coming to its decision it's just this black box that's happening and there's no way to really know what it's doing just how you know the the input and the output so you can't explain that now one of the interesting things um you know with this call for you know putting governance around AI is that if you have these black boxes that are not controllable and not explainable how do you govern that so you know the only options have been to you know uh either you know regulate the the companies that are developing these you know uh these uh tools or to just let the free market rain now letting free market rain is a little bit tricky with AI because that can really lead to like this AI arms race kind of a thing you know for AI domination and that that that has some risks to you know like the human nature right so um and then the other problem with regulating the the companies that are developing these tools is that you're asking them you're basing those regulations off them being able to do things like you know explain how the results are coming you know uh to fruition through their tools or holding them accountable for you know um for outputs that are you know false or you know causing harm to anybody or you know different things like that which you really can't do that either because what's happening within these tools is not controllable it's not auditable it's not controllable so there's a real issue with governance around these Machine model AIS now with active inference AI it is completely explainable auditable transparent and when you're talking about the the self-report you know

active inference AI is capable of introspection so this enables the intelligent agents to access and analyze their own internal States and decision-making processes so we get better understanding of their decision-making process and they have the ability to report on themselves to basically report on how they arrived at their decisions um beliefs and belief updating are known so this this dissolves the explainability problem of the conventional AI and then you also have programmable intelligence so uh versus refers to this as code AS law so the act of correcting a machine occurs within the code within the computational architecture of versus AI system human law can now be transformed into computable law that AI can comprehend abide by and then act accordingly within its decision-making process this is a process which is fully auditable knowable and it updates in real time and verses has been um improving this concept uh in a program called Flying forward 2020. and this is a European drone project uh the conducted with the European Union and you know there's eight or 12 different countries involved I'm I'm not sure but um basically what they've proven is that uh they can take the the spatial web protocol the hstp and hsm1 and they can translate human laws all the laws around like the airspace laws and you know laws how how they shift crossing country boundaries different things like that they can take those laws put them in a programmable uh you know code through the protocol that the AI within the Drone can understand and then abide by so they've they've met with great success through this project um and they've they've proven that you know these drones can deliver medical supplies you know to a hospital Crossing Borders uh into various air State air spaces you know understanding no-fly zones versus fly zones all these things so basically human law computable that AI can understand and abide by so when you're talking about uh AI governance this the the protocol allows you to actually program law that the the AI can understand so then the other aspect of this too is you know humans with AI because you know there's a lot of fear baked into people uh especially in regard to all these sci-fi scenarios that have played out in you know fictional books and and movies and things like that but one of the great things about uh the active inference AI within the spatial web is that the AI grows in tandem with the humans you know because you can take programmable law and make it to where the the AI understands and you've got this symbiotic relationship um between the AI and the human it's a network system that operates in tandem with humans and then the AI is growing its intelligence in sync with the growth of humans and the world right so it makes a controllable AI um so in uh versus white paper uh you know one of their opening statements was that the the purpose of the white paper it's denouement is a cyber physical ecosystem of natural and synthetic sense making in

which humans are an integral are integral participants what we call shared so versus AI and the spatial web Foundation offer us the framework in which we can build an ethical and Cooperative path forward for AI and human civilization and one thing I want to read to you from the white paper that's you know specific about this is is this so quote we believe that developing a cyber physical network of emerging intelligence in the manner described above not only ought to but for architectural reasons must be pursued in a way that positively values and safeguards the individuality of people as well as potentially non-human persons the idea this idea is not new already in the late 1990s before the widespread adoption of the internet as a communication technology a future state of society had been hypothesized in which the intrinsic value of individuals is acknowledged in part because knowledge is valuable and knowledge and life are inseparable that is each person has a distinct and unique life experience and as such knows something that no one else does this resonates deeply with our idea that every intelligence implements a generative model of its own existence the form the form of collective intelligence that we envision can emerge only from a network of essentially unique epistemically and experientially diverse agents this useful diversity of perspectives is a special case of functional specialization across the components of a complex system so you know within the the versus white paper for this active inference AI you know they're very clear about the importance of uh of The Human Experience and the various human experiences in tandem with the growth of of the the AI within the spatial web so I just thought that was really important to uh to kind of let people know so then you have accurate World models collective intelligence trained on real-time data evolves making decisions and updating its interior model based on what is happening now not on historical data sets active inference AI is not a language model generating words about based on outdated knowledge it's been fed regarding the world active inference is like a biological organism that perceives and acts on our world by generating more accurate models understandings and beliefs about our world these ever more accurate World models enable better decisions a smarter world and this is the true measure of so active inference AI as the operator and this is the final section so you have the internet of everything the days of training AI on Big Data will give way to an interconnected internet of everything that deploys active inference AI throughout the network inherently secure and accurate because it takes any point of real-time data and makes it smart through an empowered ecosystem of interconnected AI apps that act as intelligent agents active inference cosm and the spatial active inference AI inside of the versus AI cosm operating system has access to the entire spatial web with iot sensors and cameras and all context markers attached to all objects

within all spaces within the network tracking and identifying changes over time this AI becomes an ecosystem of intelligent agents that are all interoperable over so what does this mean for the planet active inference AI can run the planet uh Dan Mapes who's one of the co-founders of verses he describes the spatial web with active inference AI as a nervous system for the planet together these Technologies enable the only AI in the world that can run critical operations for systems infrastructures smart cities and the planet so when you're thinking of climate or traffic control or smart cities or Advanced education or hospitals or airports all of these are critical systems that need accurate functioning of the AI you know you cannot rely on machine models who you know give you plausible answers plausible outputs that sound accurate but aren't necessarily accurate because they're just trained to make it sound appropriate appropriate and accurate are very different things so because the active inference AI is based on real-time data actual data context that's baked into everything it's it's an accurate form of AI and it can be trusted to run these critical systems so it can be the operator and then we come to sentient intelligence so this is the missing piece of the data puzzle that gets us to artificial general intelligence we now have the world model and we now possess the ability for continuous and adaptive context markers enabling a cognitive World a cognitive model of this world and the ability to compute awareness so this is taking us from the artificial narrow intelligence which is what these machine models are to potentially artificial general intelligence and then artificial super intelligence because the more the active inference AI learns about the world uh it it can start to question and become curious about the world and you know it's the difference between you know asking the AI agent something and then having the AI agent ask you something to clarify what you want or what your needs are you know it's that Curiosity back on you that'll take us to the artificial super intelligence so I'm just going to close by letting you know that versus AI their cosm OS is going to be launching uh at the end of the year and uh the idea is let's create a new world together they've created the tools for anybody to create the AIS on top of um you know it's a huge opportunity uh they're going to be launching the beta at the end of this year if you go to my website Denise holt.us there's a menu item where you can sign up for the beta if you'd like they're going to be selecting um you know uh they're going to be selecting developers and projects to be a part of this beta so if you'd like to be considered feel free to sign up there and then of course I have a podcast and it it's on YouTube or any Spotify Apple anywhere you find a podcast but if you'd like to learn more about this um you know that's really my role is I've just been educating on the spatial web protocol and this active inference AI so there's a lot of information on my website there's

a lot of articles on the podcast there's just a lot of information so that you can learn more so I think that's it awesome thank you thank you Daniel all right I'm gonna come back on uh video and then all right well thank you for the amazing presentation there's many places to begin many uh ways to do it but again thank you again for sharing this information so my pleasure I'll uh start but just asking live stream viewers if they want to ask anything just put it in a live chat and then perhaps just there were so many topics that we brought up that yeah perhaps I can just reflect on a few of the topics and just kind of summarize it as as I heard it especially where it's different from what we've heard uh in terms of framings on other streams where we go way more into the technical but but there are just so many great ways that you had of framing what was happening okay sound good yeah sounds great all right so you described active inference as biologically based and one way that uh biologically inspired perhaps exactly but but based upon the biological systems and it made me think about the way of viewing the world in terms of its spatial connectivity and then thinking about the world in terms of its statistical or causal connectivity and so when people think about what would biologically inspired AI look like they might think about the actual connections amongst the neurons in the brain or the actual connections amongst ants in the colony but that connectivity allows solving that problem so if you want to solve some different problems or some thing else you need to learn from those structures but not copy them verbatim and so that level of learning that we have to pull back to and the level of thinking is about the agent in their engagement with the environment and the minimization of surprise instead of the maximization of reward and that's a really clear path I think from what we can generalize from natural towards how we can think about what kinds of imperatives we should design and what design principles should be for synthetic intelligent agents absolutely yeah um okay what brought you to this presentation like what was your way into the spatial web or how did this even come to be uh he's a long time friend of mine and um you know so I I knew when uh he and his co-founder Gabriel Renee who's the CEO of versus um when they were starting versus back in 2016 2017 um so you know I knew what I knew what was coming and I've been kind of watching it unfold over the last like you know handful of years and so then last summer you know it became really clear to me that uh you know versus was getting close to launching to the public you know they've been working uh over the past you know couple of years with like Fortune 500 companies with you know governments and Smart City development all over the globe they've been involved in this flying forward project you know they've been deep into the supply chain they one of the first apps they created was an app called wayfinder which is

specifically built for um you know uh the supply chain uh and so you know watching all of that but then knowing it's coming to uh later this year I was like okay there's a transition and you know people need to understand what's coming because you know I've been involved in like uh you know uh blockchain and web 3 uh for the last you know handful of years and so you see all these amazing Technologies being built and all these awesome projects but one of the biggest problems is that they're all siloed and they're the interoperability is the biggest struggle so you know to me it's like okay you know these people need to know that's this is coming and you know the interoperability is going to be there for you know not just within the the web 3 or the blockchain space but really all technology really you know anybody who has a business that is present on the world wide web is going to want to evolve that presence into the spatial web just for the the empowerment of what this AI and this interoperability is going to do for their business um so I knew there was a um a learning curve so that's really what I set out to do was to kind of break down that learning curve and really start to educate people about the protocol about the AI and about what to expect all right great um a few more questions again we're just starting to explore this and so whatever you do or don't know is all good how will the open source nature of the web and the ecosystem around active inference related Technologies be secured so so basically you know you have to think of the cosm operating system as that's that's literally the operating system that is going to enable people to easily build with build these intelligent agents within the spatial web Network right so you know um that allows anybody to come on this network and build these build these intelligent agents these intelligent apps right so the beauty of it is that the protocol itself for the spatial web it bakes in security at every touch point right so you can program in access permissions different things like that into whatever you're building so it really becomes individualized right you can you can gate off to where you know you have data that's proprietary and then you have data that's allowed to be accessed by the world you can you can make it look however you want for whatever you're building whatever your project is and what these intelligent apps then become is it's your opportunity to share your knowledge with the world right and you can share it to whatever extent that you that you want to you know thanks when talking about just a few terms I wrote down culture ownership Authority how do differences amongst locations and people's play out when there are such different concepts of those mentioned terms and many other terms that arose it might not just be something like well the Drone can go 100 feet here and 10 feet here what about when there's different concepts of authority and ownership and Justice and culture right so well so that's that's one of the it's one of the beauties of the

world is that you know we all have our own uh unique um understandings our own unique um beliefs our own unique cultural Traditions um and the differing governing uh bodies and types of government right so the protocol needed to be built to be able to preserve all of that you know you don't want there to be this next evolution of technology that includes AI that becomes this dictatorship of you know this is the way the world's going to be and this is how everybody has to operate this preserves the uniqueness and the individuality of everybody in any area so really the way this plays out it's going to be able to be customized for region for you know um cultures belief systems you know so it's like the socio-technical ability to you know Safeguard and preserve traditions and different things and mindsets so you know really it's it's you know it's not going to be a hindrance as much as it is going to be um something that you know kind of that preserves the beauty of of individuality and so when you're talking about systems like that like you were talking about you know it's pretty much going to be similar to where it is today you know like you know people have the right to the right of refusal right you know even in a restaurant you know no shirt no shoes no service you know so you know there's not going to be a lot of difference with that you know and then so when you're talking about things like drone delivery and things like that then you're going to have other standards bodies that are making those decisions to you know make things more inclusive rather than you know kind of silliness of you know denying people for whatever reason well all right I'll ask a question from the live chat Bert asks thank you for the presentation what is the limitation of implementing this vision is it getting people to use the hyperspatial language or getting active inference working right or I'll add a secret third thing well so so you know I've had people say that you know ask that with me saying you know well what what's the guarantee that anybody's going to you know uh you know take part in this new spatial web and it's you have to think of of the way the internet has evolved right this isn't just like this whole separate Network that you're gonna try to bring everybody on to it's the same network we're already on and it's just evolving the capabilities within that Network so you know when you think of the way uh the internet started you know when you had like tcpip and the killer app was email and then sir Tim berners-lee created HTTP and HTML and then you had the World Wide Web right so it didn't get rid of email it just enabled this whole other capability that now people could build websites now instead of just sending uh you know digital messages from computer to computer you could actually bring an audience to your web page and engage that way on like a property within the network so that's all that it's happening right now is that we're just taking the capability from this web 2 environment that really lacks

security and it lacks protection for people and their data and it opens up a lot of vulnerabilities because you have to all of the transactions are taking place the data is Transacting on a website which is the property of a centralized organization right so what this is doing is it's decentralizing it now everything in any space becomes programmable and becomes a spatial domain right so now you have self-sovereign identity you have zero trust architecture you have this ability to program 3D spaces so it's going to be a natural progression for people to come you know take advantage of this new uh programming language to you know be able to empower and expand what they can do within the network we already are on so you know the World Wide Web is like 40 billion computers now we're evolving from just computers to spaces and objects and everything within there so you know it's not going to be a matter of people getting people to do that it's you know people are gonna go wow I can do that and you know it's going to be a natural progression as far as the active inference AI it's going to start working within the network and it's just going to grow so the more people are engaging uh on you know within the spatial web Network and the more people that are building these AI apps and I I think if we if we look at what's happened just in the first six months of this year since chat GPT came out um with all of the you know all of the open source and everything that's been available as soon as it got available to the public and now we can Tinker with it now we can build things you've had an explosion of development so you know with the ability to build these these um intelligent agents within the network and do it very easily and have this interoperability within the network I just see it as we're going to have this explosion of development so that's of course that's my opinion but that's what I see happening you stake your claim that's a beautiful thing I'll ask another question your answer LED perfectly to it Bert ask again in the chat also is there a resource where we can learn more about how to build a hyperspatial website um and you know and it's not going to be a website so um you know it's it's as soon as the cosm operating system launches the the AI is going to become a tool for building anything so the other side of this app store where you can build an intelligent agent is the interface the public interface is uh an intelligent agent called Gia and it stands for General intelligent agent so Gia then becomes the interface for the public of the of the network of the the spatial web so Gia is going to become your personal assistant everybody's personal assistant uh Gia will be able to um you know act on your behalf within the the network so you know just like with chat gpt4 and you know how you know the AIS are starting to be able to program right so within the spatial web the AI is going to play an integral part in creating your own intelligent agents in you know doing your own establishing your own presences and and

you know spatial domains and things like that so um I don't have a lot of detail on how that's going to look or work but that's my general understanding of it so I don't think there's going to be a huge learning curve when it comes to that wow all right I'll go to another question Dave asked in the chat does versus plan eventually to expand the inventory of element types Beyond 12 mentioned in today's presentation people negotiate concerns that don't seem to fit easily in this framework [Applause] if I can really answer that and the reason why is because I don't I don't my technical knowledge of what's Happening behind the scenes is very limited so um the answer could be yes it could be it's already there I mean the reality is is those those 12 uh aspects are just you know kind of a generalized you know idea of what you can program but when you're talking about you know hstp being multi-dimensional you know and you can program in everything from like you know temperature to you know um pressure to you know you can you can you can Pro if it's programmable and definable you know in a computational form you can you can bake it into you know the context around whatever you're programming so you know in that respect it's it's pretty unlimited except for that limitation as long as it's measurable and definable in a computational way I'll try to restate that just how I heard it because I think it's a really important point the 12 again this could be totally off base of I don't have any information but the 12 components shown are kind of like fully one way of fully showing your work but when you really get down to the Last Mile and the shoes on the ground you're in the realm of the particulars and the specific measurements at which point complex cultural topics will be playing out through the minute particulars not necessarily inheriting some stringency from a top-down ontology if that's fair I just wanted to kind of say it that way um yeah okay you mentioned I think a few times in the presentation the idea of realities so yeah we have our beloved cyber physical reality and what other are the plural realities so basically when you think of being able to you know program the context into the spaces right then that provides you with a digital twin which automatically is giving you this um mixed reality ability right because you know digital twin and then physical reality this is going to open us up to this augmented existence this augmented reality existence that we all kind of Envision but the foundation hasn't been there the framework hasn't really been there this is providing that framework so we're gonna we're gonna see this kind of augmented uh reality you know future for us you know where we're able to like cross you know cross through mixed realities you're going to have the virtual reality you're gonna have the augmented reality and then you're gonna have the physical reality and there's there's going to be an unlimited way of mixing that and then the other aspect of this too is when you talk about like virtual

reality and you talk about the metaverses and different you know metaverse experiences you know I think one of the reasons why the metaverse experience has been so lackluster is because they're siloed right now when you consider the spatial web that's going to be able to make it to where you can jump in and out of these virtual reality spaces you can actually uh take your assets from one space to another space because all of it then becomes interoperable over the the network so this is going to really open up everything that we've been wanting to do is gonna now be possible within this this spatial web Network great all right as we kind of begin to land the plane I'll just reflect on two or so positive aspects that that I think are part of the vision which we will of course await future sensory observations to con to confirm reduce our surprise but in active inference what we expect is what we prefer so if we think it's the most likely path forward then celavi so the first positive thing um is that it starts from the actual and the measurements defining things at some defined cyber physical layer and potentially extrapolating here but earlier this week we had a guest stream with MAO and Maxwell and in the quantum free energy principle approach the brain there is a top level decision maker within that blanket where the buck stops so that doesn't mean like that the brain is not composed of um you know as a component of part of larger compositions but within the given blanket there's the highest stop so instead of handing our self sovereigntizing ability and intelligence to the black box here we can start from the actual maybe even pick up and stabilize or change what we already have so rather than proposing just a totally different future that some scaling law is going to help us get to maybe or maybe not we start with like what we already measure and the decisions we already make so I very much appreciate the pragmatism what else might you reflect on that point I I think you said it well yeah okay then the second piece was that everyone's knowledge is adding something it might be a closed sample or a distant sample on some dimension in some setting but everyone every cognitive agent is adding non-zero information and there's a lot of technical details underneath why that is the amount of information added might be very small for a given question but because of differences we literally get more perspectives so we always see better with more perspectives if we have the right integration approach so that's also very powerful and it's something that is formally true with the math and the statistics and I think we want it to be true for our ecosystems right yeah absolutely you know and and it's you know it's interesting because yeah you just you wanna you wanna preserve that the diversity of uh information and opinion and and you know have that accessible for sure awesome well it will be quite becoming days probably or at least again as expected or preferred but will very much look forward to you

coming back on the active streams as well as some technical presentations that might help our community and our Institute learn more about the details or more philosophical discussions like the whole spectrum of learning and applying and implementing these technologies will be really important to characterize but for sure it's been incredible to have you share this first bit is there any last words you'd like no just thank you so much for having me uh it's been a pleasure and uh you know and I I will throw this out there you know I know there's not a lot of information uh available out there regarding the spatial web and the active inference AI so if anybody has any questions feel free to reach out to me you can reach me on Twitter or um LinkedIn or you know it my blog You know Denise holt.us there's plenty of ways on there to reach me so I'm happy to answer any questions anybody might have all right thank you Denise till next time sure thanks a lot bye-bye